

STAT 5000

Statistical Methods and Application I

FORMAT	CATEGORY	CREDITS
On-Campus	Core	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 1 – Statistics Pathway (on-campus)	Machine Learning Engineer (62.3 / 100)	43.6 / 100
SPECIALIZATION Applied Mathematics Pathway		

44

CURRENT AVG SCORE
across 6 job roles

84

POTENTIAL SCORE
with recommended additions

+45

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces exploratory data analysis, probability theory, statistical inference, and data modeling. Topics include discrete and continuous probability distributions, expectation, laws of large numbers, central limit theorem, statistical parameter estimation, hypothesis testing, and regression analysis. Considerable emphasis on applications in the R programming language.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	62.3	100.0	+37.7	HIGH
ML Engineer	45.5	90.1	+44.6	HIGH
Data Scientist	42.4	80.2	+37.8	HIGH
Applied Scientist	41.9	82.7	+40.8	HIGH
AI Engineer	38.8	84.1	+45.3	HIGH
MLOps Engineer	30.8	67.6	+36.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

62.3 → 100.0 (+37.7 pts)

predictive sets learning tools business design development science

ML Engineer

45.5 → 90.1 (+44.6 pts)

learning systems models model build problems technical terms

Data Scientist

42.4 → 80.2 (+37.8 pts)

rules sets build problems

Applied Scientist

41.9 → 82.7 (+40.8 pts)

storage problems models technical learning systems design development

AI Engineer

38.8 → 84.1 (+45.3 pts)

models learning build software technical model systems based

MLOps Engineer

30.8 → 67.6 (+36.7 pts)

model systems learning tools models development error

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥ 200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.